Game systems

Progression games



Bootcamp overview

- The core components of a game economy.
- How to balance these elements to create a fun and fair experience.
- An introduction to Machinations, a powerful tool for modeling and testing these systems.

Progression games

Progression games are designed to give players a powerful sense of accomplishment. They follow a clear cycle to keep you engaged:

- Complete a goal (quest or a level)
- You get a reward, such as a new ability or a powerful item.
- You feel more powerful, which motivates you to continue playing and face new challenges.

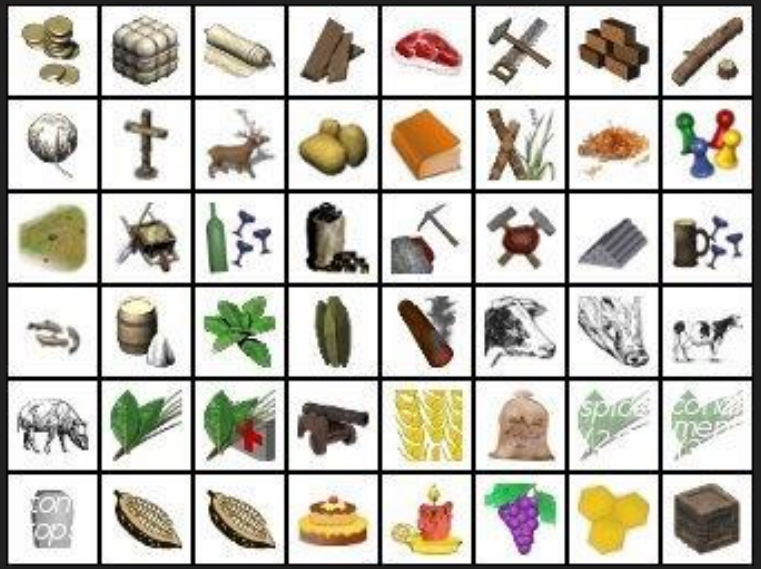


What is a Game Economy?

At its core, a game's economy is all about resources and how they relate to each other.

Resources are any concept in a game that can be measured with numbers. This can be a tangible thing like gold, wood, or diamonds, or something more abstract like a character's health or a strategic position on a map.

Anything that can be measured is a resource. But just because a resource exists doesn't mean it's useful or even used in your game.

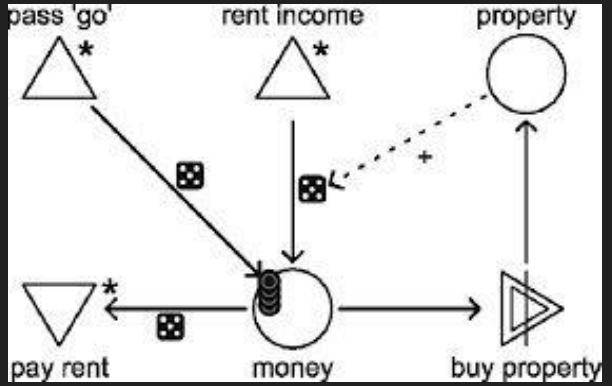


Why Study Game Economies?

By looking at your game's resources and how they flow, you can build a game that is fun to play and feels fair.

- Balance & Fairness: Spot and fix problems that make the game feel unfair, like a boss that's too hard to defeat or an item that's too expensive.
- Determine Goals: Figure out what players should be aiming for, from big goals to small, satisfying sub-goals.
- Elevate Mechanics: Make your core game loops feel more meaningful and rewarding.
- Improve the Game: Identify weak spots and make your overall game experience better.

To do this, you'll need to use a lot of math! Tools like Excel or specialized software like Machinations are a huge help.



Doom

In Doom, what resources can we identify?



Doom

In Doom, what resources can we identify?

- Ammo
- Health
- Armor
- Enemies on the Screen
- Stage in the Game



How about Brawl Stars?



How about Brawl Stars?

- Health
- Ammo / Basic Attack
- Super Ability
- Power Cubes
- Trophies
- Coins & Gems



Minecraft



Minecraft

- Materials (Wood, Stone, Gold, Diamond)
- Hunger & Health
- Time of Day
- Experience Points (XP) / Levels
- Speed (via potions and enchantments)
- Enemies on the Screen



Tangible vs. Intangible Resources



Resources in a game can be split into two main types:

- Tangible Resources
  -These have a physical presence.
  -They exist as an object in the game world that a player can interact with directly.
  Think of a health potion on the ground or a piece of wood from a tree.
- Intangible Resources
  -These don't have a physical form in the world.
  -They are typically displayed in the game's UI, like in an inventory or on the HUD.
  Think of your player's experience points, mana, or a score.

Concrete vs. Abstract Resources

Besides being tangible or intangible, resources can also be either concrete or abstract.



Concrete Resources:

- These are clearly visible and recognized by both players and the game's systems.
- They are the things you can point to on the screen and say, "I have 5 gold," or "That is a health pack."
- Their value is explicit and tied to a rule or number in the game.

Abstract Resources:

- These don't "exist" as an object in the game world, and they aren't tracked with a visible number in the UI.
- They are assets that you use within the game's systems and mechanics to gain an advantage.

What is an Abstract Resource?



An abstract resource isn't part of the game's official rules or user interface. It's a valuable asset that players create and use to gain a hidden advantage.

In a Tower Defense game, the maze you build is an abstract resource. You use your strategic knowledge to create a structure that slows down enemies, giving you the valuable resource of time to win.

Another example is teaming up with other players. Trust and communication with your teammates are abstract resources. They don't have a value on your HUD, but they are crucial for success in multiplayer games. For example, in a cooperative game, a well-coordinated team can take down a powerful enemy far more effectively than a group of uncoordinated players, even if their characters have the same stats.

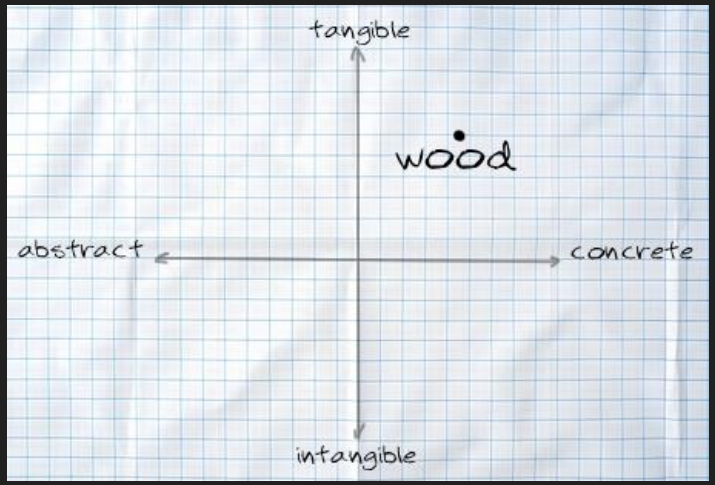
Exercise

Resources are key ingredients to shape gameplay. By analyzing them, you'll gain a deeper understanding of how a game truly works.

Your task:

- Choose a game: Think of a game you know well.
- Make a list: List at least ten different resources from that game. Try to pick resources that serve different purposes.
- Plot your resources: Create an axis with 'Concrete' and 'Abstract' on the horizontal axis and 'Tangible' and 'Intangible' on the vertical axis. Then, place all your resources on this chart.
- The Big Question: Which single resource is essential to achieving victory? Explain why.

This exercise will help you think like a systems designer and uncover the hidden power of a game's economy.



Victory resources

A victory resource is the core resource that a player must acquire or manipulate to achieve the winning condition. It represents the ultimate goal.

In some games, winning is as simple as collecting the victory resources. For example, in a game where the first player to collect five diamonds wins, the diamonds are the victory resource.

In other games, the goal is more complex. You might win by defeating a final boss, which is effectively reducing the boss's HP resource to zero. In this case, the boss's HP is a resource you must deplete to achieve victory.

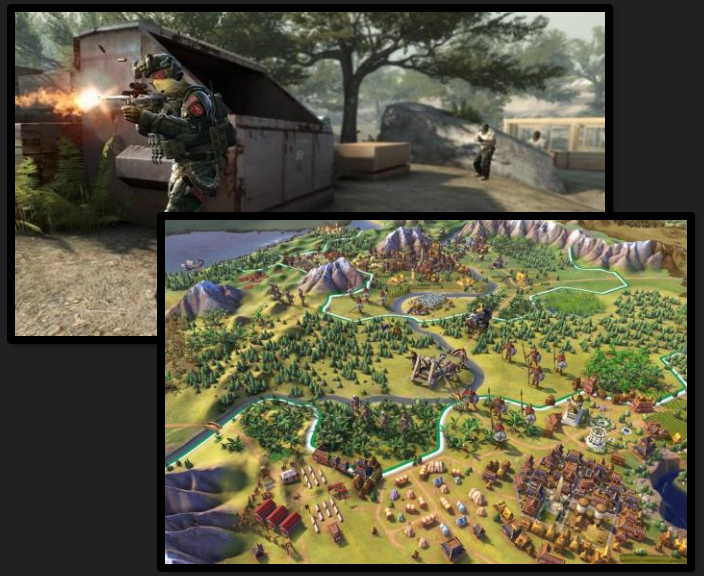


Mechanics and the Victory Resource

It's a great exercise to break down a game's objectives to find its Victory Resource. Many different mechanics can be used to achieve this ultimate goal.

Killing can be a path to victory, as seen in Counter-Strike. The final win condition is achieved by eliminating the enemy team's players, which can be seen as depleting their collective "life resource."

Building is another path to victory, like in Civilization. Here, you win by creating and managing resources to build a powerful empire or complete a scientific breakthrough.



Resources Influence Each Other

Resources don't exist in a vacuum. They influence each other in ways that can change your game and how players interact with it.

- Upgrades and Power-Ups: Collecting one resource, like XP or gold, lets you improve another, like your attack power. This fundamentally changes how you approach challenges in the game.
- Unlocking New Content: An example can be found in Super Mario 64. Collecting stars (a concrete resource) unlocks new stages and sometimes even new stars in stages you've already visited. This shows how collecting one resource can create a chain reaction that expands the entire game world for the player.

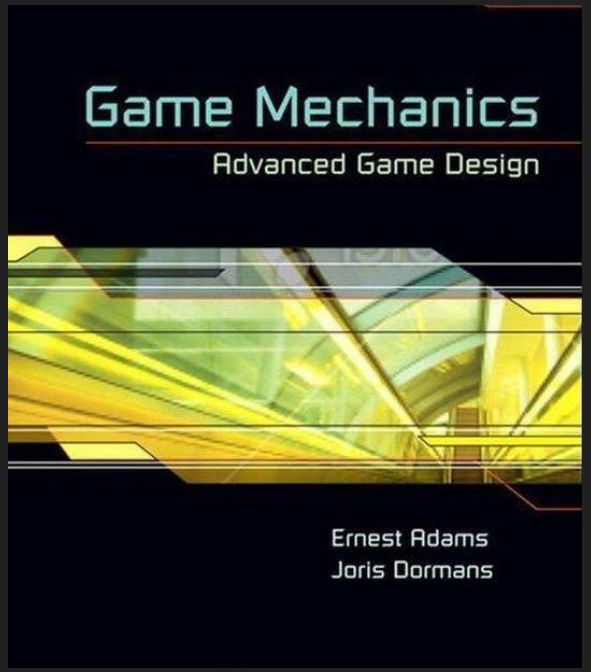


The Four Game Economy Mechanisms

All player interactions with resources can be categorized into four fundamental mechanisms:

- Sources: The creation of resources.
- Drains: The consumption of resources.
- Converters: The transformation of one resource into another.
- Traders: The exchange of resources between players.

These mechanisms form the foundation of our discussion, as they are the core concepts used by Ernest Adams and Joris Dormans in their book The Game Design Handbook: A Unified Approach to Designing Interactive Experiences. This lecture is based on their work and the principles they use to explain video game economies.



Sources and Drains

Sources

- Mechanisms that generate resources out of nothing
- Can be timed or activated upon a certain condition
- Ruled by a Production Rate

Drains

- Mechanisms that destroy resources completely
- Can be timed or activated upon a certain condition
- Ruled by a Destruction Rate



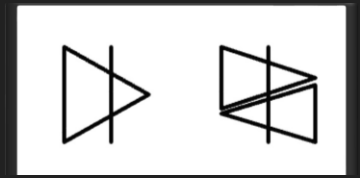
Converters and Traders

Converters

- Mechanisms that transform a resource of one type into another
- Mix of a Source and a Drain
- Ruled by a Conversion Rate

Traders

- Mechanisms that exchange resources into different entities
- Resources are neither destroyed nor created
- Ruled by a Trade Rate



Stardew Valley: Sources and Drains

Stardew Valley is an example of a progression game with a clear and effective economy. Here's how it works using the four core mechanisms:

Sources: How are resources created?

- Farming: Crops grow and provide produce.
- Foraging: You find wild mushrooms or berries.
- Mining: You gather stone and ores from the mines.
- Crafting: Your workbench creates new items.

Drains: How are resources consumed?

- Energy: Most actions (like farming or mining) drain your stamina.
- Health: Getting hit by enemies drains your health.
- Spending: You spend money at Pierre's shop.



Stardew Valley: Converters and Traders

Converters: How are resources transformed?

- Cooking: Raw ingredients become cooked meals that restore energy.
- Crafting: Wood and stone become tools or furniture.
- Selling: Any item can be sold to become money.

Traders: How are resources exchanged?

- The primary exchange in the game is between the player and the in-game shops, which is the core loop of this economy.

By analyzing the game this way, you can see how the designers balanced sources and drains to create a satisfying loop of progression.



Resources are Independent from Mechanisms

This is a key concept: a resource itself is separate from the mechanism that uses it.

Any single resource can be used by all four mechanisms in a game's economy. Let's use cauliflower as an example:

- Source: You can receive a cauliflower from a mysterious package in the mail (a new resource is created out of nowhere).
- Drain: You can throw a cauliflower in the trash (destroying the resource).
- Converter: You can cook a cauliflower to create a meal (converting the raw resource into a cooked meal).
- Trader: You can sell a cauliflower at a shop to get money (exchanging it for another resource).

This shows the flexibility of each resource and how its purpose depends on the game's mechanics.



The Core Loop: The Heartbeat of Your Game



The Core Loop: The Heartbeat of Your Game

Every game has a Core Loop—the main cycle of actions players continuously perform. Think of it as the game's heartbeat: it keeps everything moving and keeps players engaged.

A Simple, Powerful Cycle

- Effort: The loop starts with a basic action, like gathering resources or defeating an enemy.
- Reward: This action leads to a reward, like getting gold or leveling up.
- Progression: The reward then makes the initial action more effective or unlocks new challenges, encouraging you to repeat the loop.

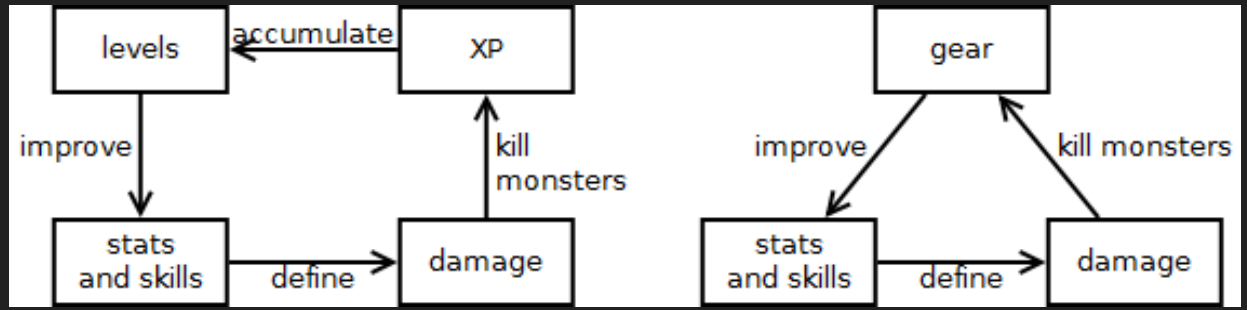
A strong core loop is what makes players stick around. It creates a satisfying rhythm of effort and reward that keeps them coming back for more.

Weapon upgrades

This simple loop is the engine behind many action games.

- Effort: You kill enemies to earn a reward.
- Reward: Killing enemies gives you loot, which you then use to get better weapons.
- Progression: The better weapons make it easier to kill more enemies, which gets you more loot, and the cycle continues.

This loop creates a continuous rhythm of challenge and reward that keeps the player engaged and constantly progressing.

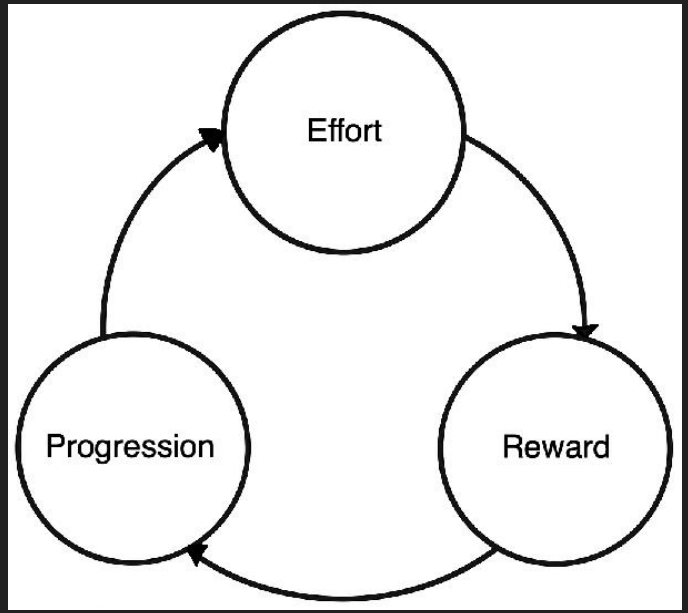


Your Turn: Find the Core Loop

Think about a game you know well.

Try to describe its core loop in just a few sentences, using the Effort > Reward > Progression model we just discussed.

- What is the effort? What action do you repeat? (e.g., clicking, moving, killing enemies)
- What is the reward? What do you get for that action? (e.g., gold, XP, new items)
- What is the progression? How does that reward help you with the next action? (e.g., get stronger, unlock new areas)



Balancing the core loop: Machinations

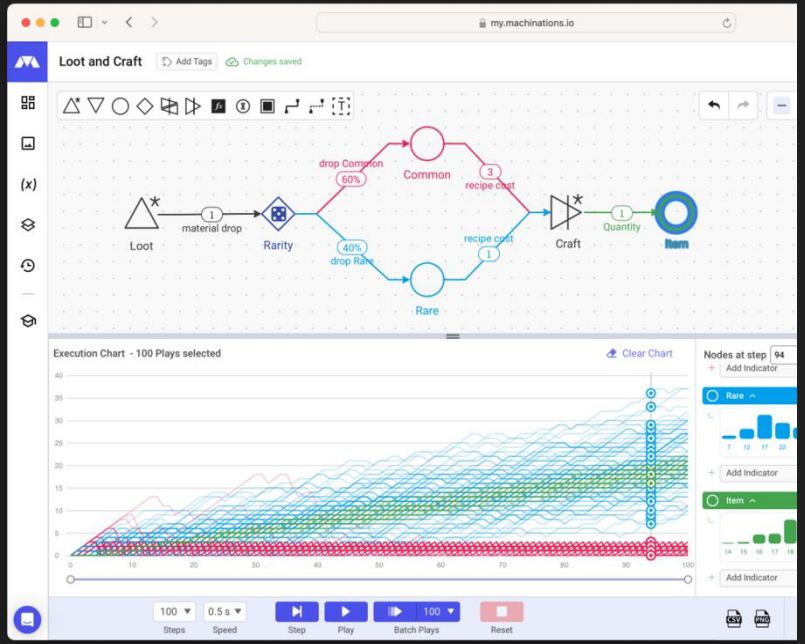
Machinations.io is a web-based tool for game design and balancing.

It lets you visualize and test game systems like economy, resources, and player progression.

You can create diagrams to see how points, rewards, and actions interact in a game.

It's useful for students, designers, and developers to experiment safely before coding.

Find the software on machinations.io

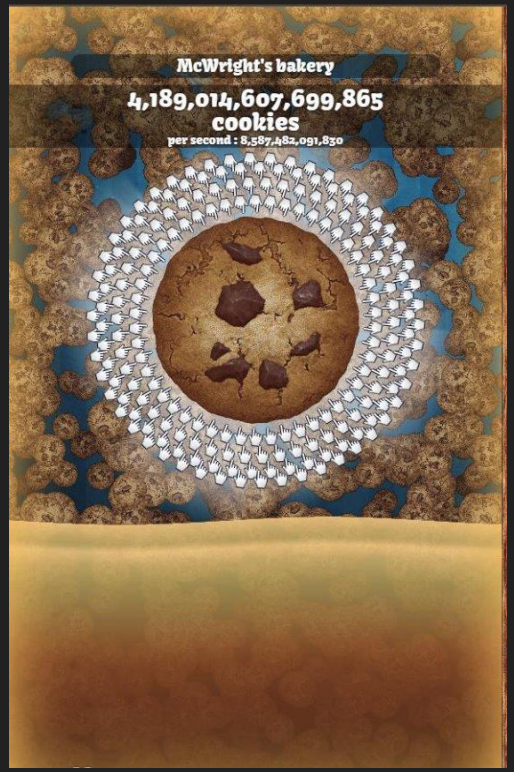


Subgenre: incremental games

And interesting subgenre are Incremental games (also known as "clicker games" or "idle games") They are a type of progression game focused on exponential growth. They follow a simple loop that rewards minimal effort:

- You perform a simple action (like clicking a button) to earn in-game currency.
- You spend the currency on upgrades that automate the process and generate more currency for you.
- The rewards increase exponentially, allowing you to unlock more powerful upgrades and feel a constant sense of progression.

This cycle is designed to make players feel productive, even when they're not actively playing.



Reputation

Despite their popularity, incremental games often face criticism, largely due to:

- Perceived Lack of Skill: Many players don't consider them "real games" because they require minimal skill beyond clicking or waiting.
- Aggressive Monetization: The genre often relies heavily on microtransactions, allowing players to pay to skip waiting or accelerate progress, which can feel exploitative.
- Addictive Loops: The constant flow of small rewards and the infinite progression can create a highly engaging, sometimes addictive, loop that some find problematic.

However, the core psychology of their "progression loop" is also a powerful tool, subtly influencing many other successful game genres.

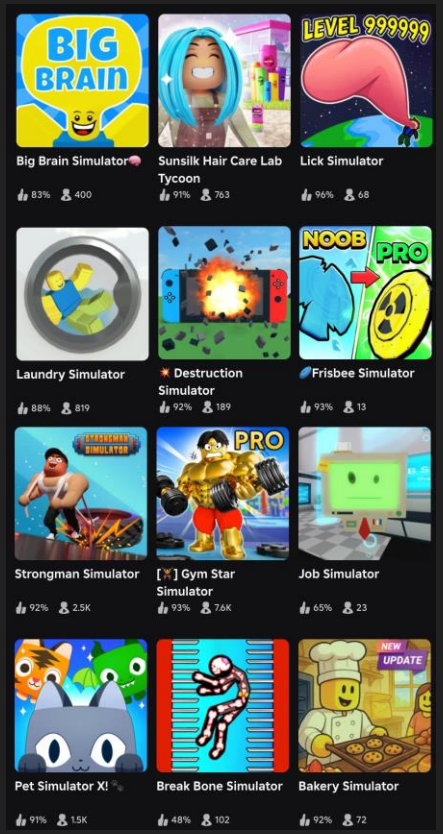


"Simulator" genre

Within the Roblox ecosystem, simulator games are a unique genre that took the single-player incremental game formula and made it a multiplayer phenomenon. While most incremental games are solitary experiences, this genre thrives on a competitive social loop:

Players perform a simple action to earn resources, which they then use to grow more powerful.
This growth isn't just about personal achievement; it's a race against the other players.
The real drive comes from competing for a high rank on leaderboards and showing off your progress with rare items and pets.

The genre is one of the most popular and commercially successful on the platform, often attracting hundreds of thousands of players at once.



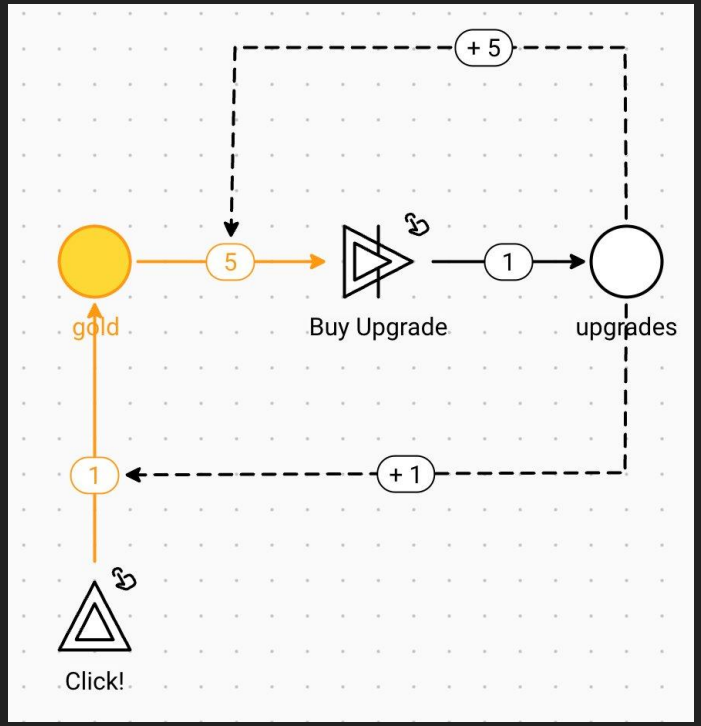
The power of progression games: focus, feedback, systems

Our assignment was based on these two-player progressional games.

- Focus on the Core: The simple rules of progression games allow you to concentrate on what truly makes a game engaging: building a well-tuned system that is both fun and fair.
- Direct Balance Feedback: In a two-player setup, imbalances become immediately clear. This forces you to analyze and adjust your design to ensure fairness, and allows you to quickly test and iterate on your design.
- Systems Thinking in Action: You'll learn how small changes to game variables can significantly impact the overall player experience, revealing how every part of a game is connected.

This exercise will equip you with essential skills for designing effective and enjoyable game systems.

Example: cookie clicker



This diagram represents a simple cookie clicker system.

It starts with an Interactive Source that generates resources and feeds into a main Pool. From this Pool, the player can use a Converter to purchase upgrades in a secondary Pool. Each upgrade increases two things:

- The price of the next upgrade by +5 (starting at 5).
- The output of the Interactive Source by +1 per upgrade.

This setup demonstrates a classic incremental/idle game mechanic: the more upgrades you buy, the more resources you generate, but each upgrade becomes progressively more expensive.

The diagram visually shows how the Interactive Source, Pool, Converter, and Upgrade Pool interact to create a self-reinforcing growth loop.

A Case Study: The Flower Game

Let's explore game economy concepts using a simple, conceptual game.



The Game:

- Players can buy flower seeds.
- They can plant these seeds.
- After the flowers have grown, they can be sold.

The Goal:

- The player's objective is to pay off a debt of 100g to win the game.

By focusing on this simple game, we can see how complex systems arise from just a few basic rules.

The Iris: Values and Profit

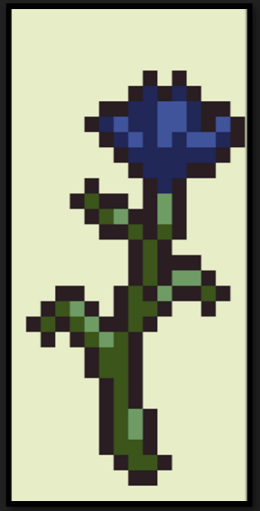
To build a balanced game economy, we first need to establish the values for our resources. Currently, we have a single resource: the Iris.

- Initial Values: The player starts with 3 seeds.
- Cost: An Iris seed costs 10 gold.
- Sale Price: An Iris flower sells for 12 gold.
- Time to Grow: An Iris flower takes 5 turns to grow.

Why these numbers matter:

- Profit: The core driver of the game's economy is the profit: Sale Price - Cost = 2g.
- Calculations: To reach our goal of 100g, a player needs to sell about 50 Irises. If we would start the game without having irises, it would take 250 turns.

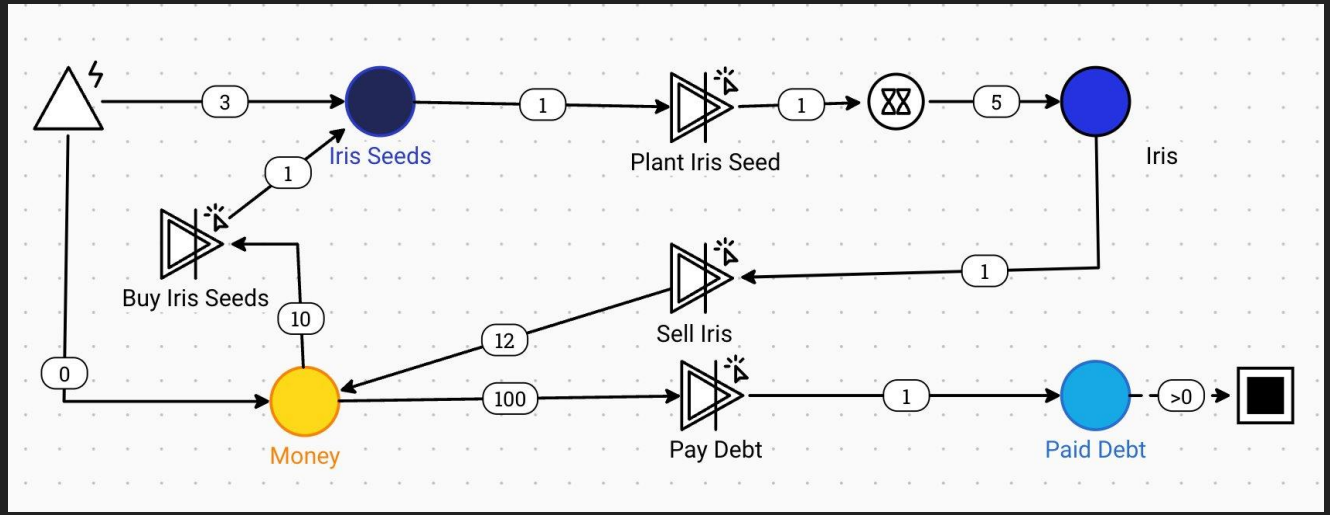
This is a great first step in understanding the mathematical side of game design.





https://tinyurl.com/iris-only

Machinations Diagram with the current game



This diagram visualizes our Flower Game.

The game rules are as follows:

- Buying Seeds: Players start with 3 seeds and can buy more seeds for 10 gold each.
- Planting & Growing: You can plant one seed at a time. The sand timer icon in the diagram represents a 5-turn delay for the flower to grow.
- Harvesting: Once a flower has grown, you can harvest it and sell it for 12 gold.
- Winning: The game ends once you have paid off your debt of 100g.

See if you can beat the game and pay off your debt by following these simple rules.

Simulating: The Iris Only Version

If we were to simulate this game with only the values we've discussed, the results would look similar to this chart.

This version of the game is quite "well-behaved" and seems rather monotonous.

- Lack of Options: The game lacks a sense of strategy because the player has only one path to victory: planting, growing, and selling irises.
- Monotonous Gameplay: There's a bit of strategy in how you spend your initial gold, but after that, it's a predictable loop without any meaningful alternatives.

This simulation shows why a simple core loop, while functional, isn't enough to keep players engaged. A good game needs more than just a single, linear path to victory.



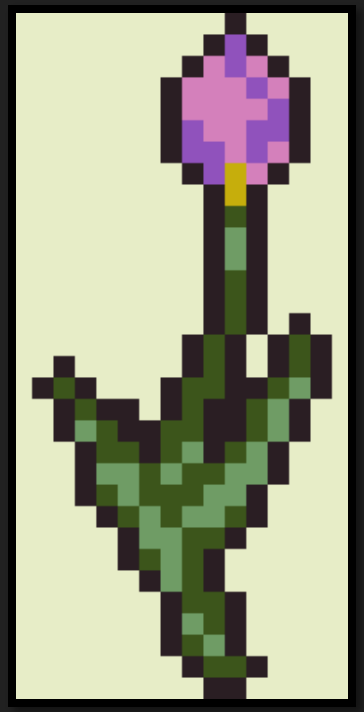
Adding a New Resource: The Tulip

Now, let's add a new flower to our game: the Tulip. The values for the Tulip are deliberately different from the Iris.

- Cost: A Tulip bulb costs 8 gold to buy.
- Sale Price: A Tulip flower sells for 16 gold.
- Time to Grow: A Tulip takes 10 turns to grow.

Why add a new flower?

- By introducing the Tulip, we're adding a strategic choice for the player. The Tulip has a higher profit margin (8 gold vs. 2 gold), but it takes twice as long to grow.
- This forces the player to make a trade-off: do I go for a quick, low-profit flower (the Iris), or a slow, high-profit flower (the Tulip)?
- This is the basis of a more engaging game economy.

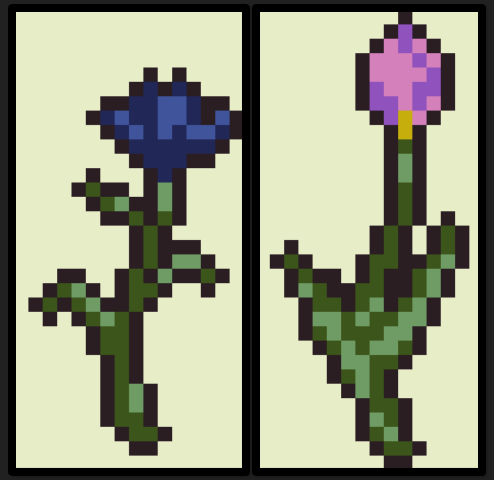




https://tinyurl.com/iris-tulips

Machinations Diagram with the current game

Profit Per Turn: The True Measure of Value



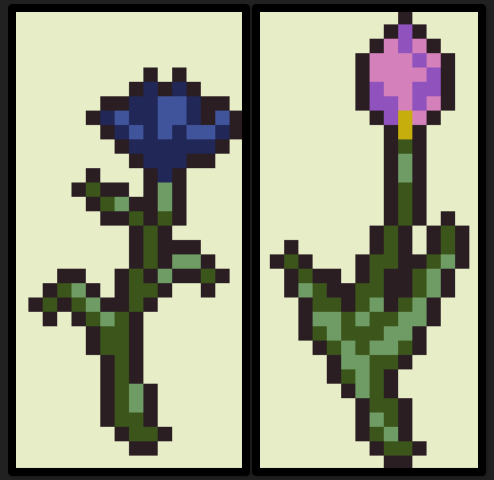
A good way to evaluate a resource is to look at its value over time.

This approach helps us make smarter design decisions. We can use a simple formula to calculate the profit per turn:

$$Profit\ Per\ Turn = \frac{Sale\ Price - Cost}{Turns\ to\ Grow}$$

This formula is a key tool for comparing different resources and ensuring your game is well-balanced.

Comparing Profit: Iris vs. Tulip



Now that we know how to calculate profit over time, let's compare our two resources.

Iris Profit:

- Profit: 2 gold
- Time: 5 turns
- Profit per turn: 2 gold / 5 turns = 0.4 gold per turn

Tulip Profit:

- Profit: 8 gold
- Time: 10 turns
- Profit per turn: 8 gold / 10 turns = 0.8 gold per turn

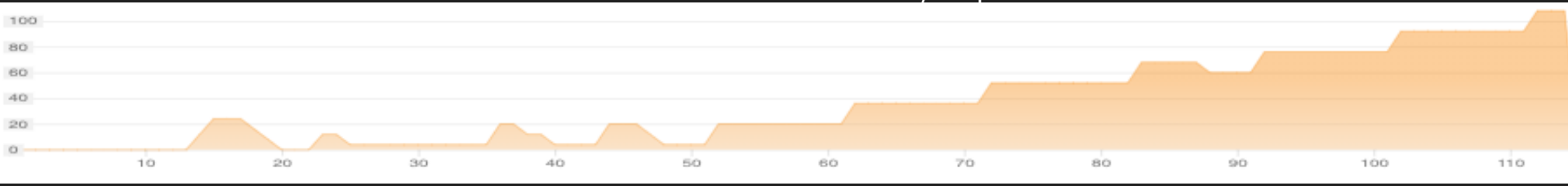
The calculations show that the Tulip is a more profitable strategy in the long run. The higher time investment per turn for the Tulip is rewarded with a higher profit per turn. This trade-off between profit and time is a fundamental design choice that gives players a meaningful decision to make.

The Game is Unbalanced, but...

The Game is Unbalanced, but...
Based on our calculations, we can clearly see that a strategy of only using Tulips is better than using Irises.

But this is not necessarily clear to the player at the start of the game, and they might still get a lot of enjoyment out of the game while still having options.

Simulation with only Tulips



Simulation using both Flowers



Balancing the Values: An Example



Balancing the Values: An Example

If we want to balance the game, we can change the values of the Tulip to bring its profit per turn in line with the Iris. Remember, the Iris gives us 0.4 gold per turn, while the initial Tulip gives us 0.8 gold per turn.

Here are three ways we could change the Tulip's values to make it a more strategic choice:

- Increase growing time to 20 turns.

(16 gold - 8 gold) / 20 turns = 0.4 gold per turn

- Increase the seed cost to 12 gold.

(16 gold - 12 gold) / 10 turns = 0.4 gold per turn

- Reduce the sale price of a Tulip flower to 12 gold.

(12 gold - 8 gold) / 10 turns = 0.4 gold per turn

By adjusting just one of these values, we can make the game balanced and give the player a real, meaningful choice to make.

Perfect balance is boring

Our last exercise showed us how to achieve perfect balance, where all strategies have the same value. But the problem with this approach is that it leaves the player with no meaningful options.

Both alternatives are virtually identical in terms of profit per turn. This can make the gameplay feel monotonous. A good game needs meaningful choices that require the player to think about their approach and risk tolerance.



As stated by Jeff Kaplan, former Overwatch's Game Director: "The perception of balance is more powerful than balance itself." (link)

The Power of Chance: Re-evaluating the Iris

Let's re-evaluate the Iris to make the game more exciting.

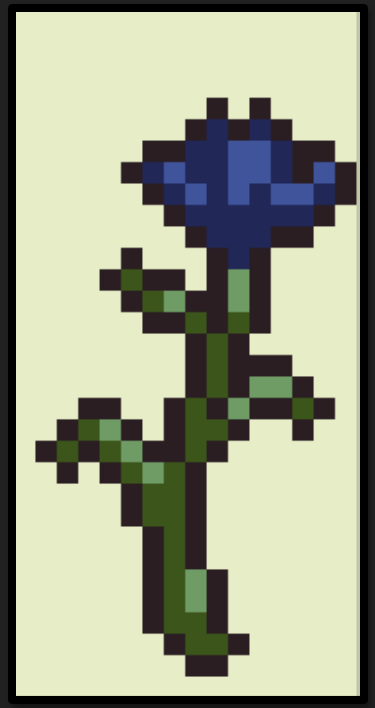
We'll add a new rule: when a player harvests an Iris flower, there is a 40% chance it will generate a new Iris Seed.

This change has a major impact on our game economy:

It adds an element of risk and excitement to the gameplay. You never know if you'll get a free seed.

This new random source for seeds can help players avoid a deadlock. Even if you run out of gold, you have a chance to continue playing.

It transforms the game from a predictable, mathematical system into a more engaging one.



The Power of Chance

Due to the new mechanic we added, the Iris is now a much more viable option. Just using Tulips is no longer the only winning strategy.

It's a matter of choice: Do you strictly follow the numbers with the Tulip? Or do you take a chance with the Iris and hope for a lucky outcome?

Balance is not perfect: The game is no longer perfectly balanced, and that's a good thing! The introduction of chance and risk creates a more engaging experience for the player.

This is a great example of how a small change in game design can completely change the way a player thinks and plays.

Mechanics, Balancing, and Fairness

It's very difficult to reach a state of perfect equilibrium in a game. Instead of aiming for perfect balance, it's often better to aim for a feeling of fairness.

- Fairness means that the player feels the game is just.
- The player feels that different options are viable, and that they can ultimately lead to a victory.
- By giving players more options, you not only make the game fairer but also increase its replay value.

Adding more options and mechanics means more work to keep the game balanced, but it's a worthwhile trade-off for a more engaging and fun experience.

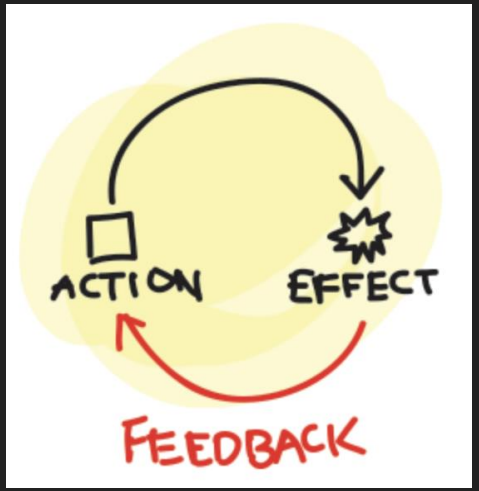


What are Feedback Loops?

A feedback loop is a system where the result of an action is fed back into the system, creating a continuous cycle.

There are two main types of loops:

- Positive Feedback Loop: An effect that amplifies itself. Think of a viral meme: a few shares lead to more visibility, which leads to even more shares. The effect grows exponentially.
- Negative Feedback Loop: An effect that gradually dampens itself. An effect that gradually dampens itself. For example, the crowd at a festival. When it gets too busy, people start leaving, which causes the crowd to shrink to a more manageable size.

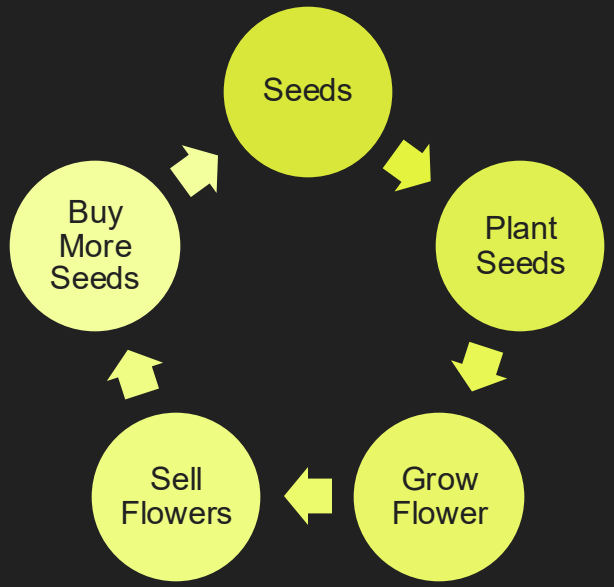


Positive Feedback Loops

In games, we use the powerful concept of a positive feedback loop to empower our players. When you're doing well in a game, you often get rewards that make you feel stronger. These rewards then help you do even better, creating a satisfying sense of momentum.

In our Flower Game, we could think of it like this:

- Action: You sell a flower.
- Reward: You receive gold.
- Empowerment: You use the gold to buy a better shovel or watering can.



Positive Feedback Loop: The Rich Get Richer

Let's try adding a new mechanic to the game that works as a positive feedback loop.

You can dissect Irises to improve your chances of getting a new seed. Instead of just selling the flower for gold, you can convert an Iris into a +5% chance to get a new seed when you harvest.

Result: You can now harvest flowers even faster, earning even more gold and feeling a satisfying sense of progress.

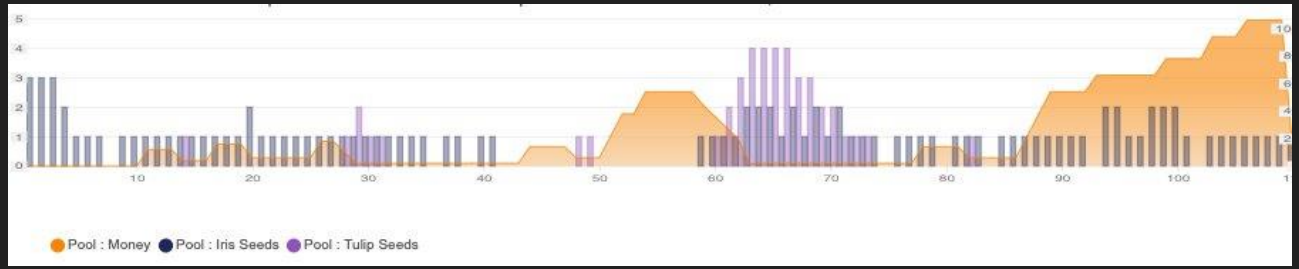


Balancing positive feedback Loops

Positive feedback loops are great for empowering players and expanding strategic choices. But they can also quickly destabilize a game.

- After handing in 12 Irises, each Iris has a 100% chance of generating a new free seed. At this point, the game is broken. The player has an infinite resource.
- The game's goals might limit the effect of a positive feedback loop, but this needs to be a deliberate design decision, not a side effect.

This shows that while positive feedback can be a powerful tool, it needs to be carefully managed to prevent the game from becoming unbalanced.



Negative feedback loops and downgrades

In games, we use negative feedback loops to create balance and prevent a single strategy from becoming too dominant. It's a way for a mechanic to become "a victim to its own success".

In our Flower Game, we could introduce a negative feedback loop to the Tulip:
Action: You are using a successful strategy, like planting and selling a lot of Tulips.
Result: The game responds by making that strategy less effective.
Dampening: The more Tulips you sell, the less gold they are worth.

This forces you to consider other strategies, like planting Irises, which is crucial for a fair and engaging game.



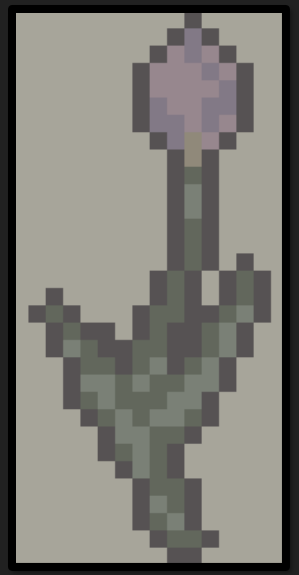
Negative feedback loop: The downside of success

Inspired by the historical "Tulip Mania" in the Netherlands, we're going to add a new mechanic that works as a negative feedback loop.

- Every time a Tulip is sold, the value of the next one is reduced by 2 gold (from 16, 14, 12, ..., to 0). Although they have a high initial value, they devalue over time.
- Dampening: The value of Tulips can be restored by +2 gold for every other flower (like an Iris) that is sold.

Result: This forces a "market" diversification. A viable strategy could be to sell one of each flower to keep the value high.

This mechanic prevents the Tulip from becoming a single, dominant strategy. It makes the game more dynamic and requires the player to constantly adapt their approach.

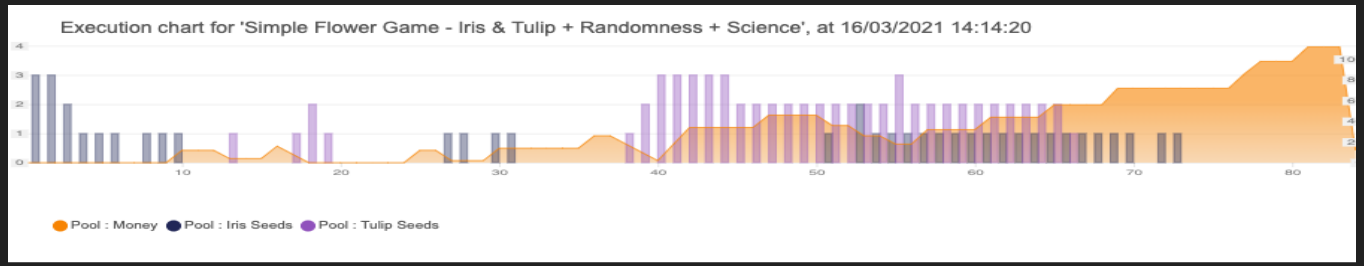


Balancing negative feedback Loops

Negative feedback loops weaken relations to counter the growth caused by positive feedback. They tend to stabilize the game over time.

- After selling 3 consecutive Tulips, their value is reduced to 10 gold, which results in a lower profit than regular Irises (0.375g for Tulips, compared to 0.4g from Irises).
- This process helps to keep the game goal fair and maintains the sense of competition. All options (and players) have a reasonably good chance to win the game.

This shows that negative feedback is a vital tool for ensuring that all strategies remain viable and the game stays balanced and fun.





https://tinyurl.com/iris-flower-game

Balancing

Finding the right balance is a process of trial and error. You can visualize your economy using spreadsheets or specific tools like Machinations before you even build the game.



When you are developing, experiment with the 'magic number' in Machinations and find the value that best fits your game. By modeling and testing, you'll discover the perfect balance that will keep your players coming back for more.

https://www.youtube.com/watch?v=Lu-RjxeDpU8

Design Start: Your Core Loop



Start your design by thinking about the core loop of your game. Think about the primary cycle of actions that players will perform repeatedly. The core loop is the engine of your game, consisting of these four steps:

- Effort: What action does the player perform?
- Reward: What reward does the player receive from that effort?
- Progression: How does the reward enable the player to repeat the initial action more effectively, or take on new, greater challenges?
- Investment: How can the player invest a portion of their reward in an upgrade or new ability?

Make sure the game has a satisfying rhythm of effort and reward, so players stay motivated and engaged.

Integrate Your Theme Into the Game

Your game's theme is more than just a backdrop. It should shape your rules and mechanics. The numbers and systems you design must make sense within your chosen theme.



- Logical Rules: Your rules should reflect your theme's logic. For example, in a pirate game, you might collect gold coins to get a new boat. The upgrade you buy for your boat (like a new sail) should let you collect gold coins faster.
- The System as the Challenge: The game's system is the main challenge. Whether you're growing a magical garden or building a factory, the goal is to understand and optimize the system you've created.
- Create a Player Fantasy: Your theme should give the player a sense of power. Your design should make the player feel rewarded by seeing a complex system grow and succeed because of their choices.

Preventing Deadlocks

A deadlock is when a player is completely stuck because they lack a specific resource needed to progress. It's a critical failure in game design, as it prevents any further action and brings the game to a halt.

If a player spends all their money on a single type of flower and its value plummets, they may end up without resources to buy anything else, effectively breaking the game.

The developers of Plants vs. Zombies avoided this by providing a reliable, source of resources: the falling sunlight. This ensures a constant but slow stream of resources, so players always have a path forward.

By offering alternative mechanics or free resources, you can ensure players always have a path forward.



Randomness is a friend

As we saw, many of these economic relations can be mapped to mathematical equations and formulas.

Ultimately, a "perfect" strategy could be calculated using the available data (and testing), which tends to create a Meta Game. When game mechanics are not clearly deterministic, these strategies can fall short.

Random elements can allow different strategies to be viable, change the game's course, and add risk, excitement, and engagement to the game.



Beyond the Math: Motivating and Storytelling



Regardless of the mathematical discussions, the game economy is directly related to the game's mechanics. It's a powerful tool that can be used to:

- Motivate players
- Empower players
- Give them options
- And act as a storytelling tool, by activating specific mechanics or justifying their purpose.

For example, a sudden drop in a flower's value can be seen not just as a mathematical consequence, but as a narrative event. As players progress, they might unlock new flowers, with new relations and new stories, thereby expanding the game's economy and narrative simultaneously.

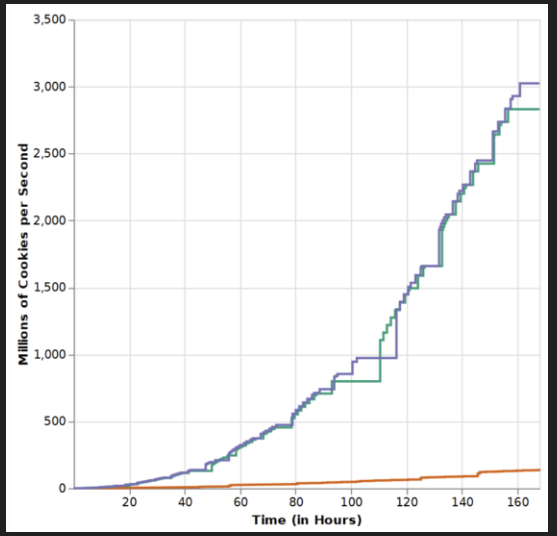
Finding the right values

Start by studying successful games in the same genre. By analyzing their economies, you can discover the unwritten rules that have already been proven to work.

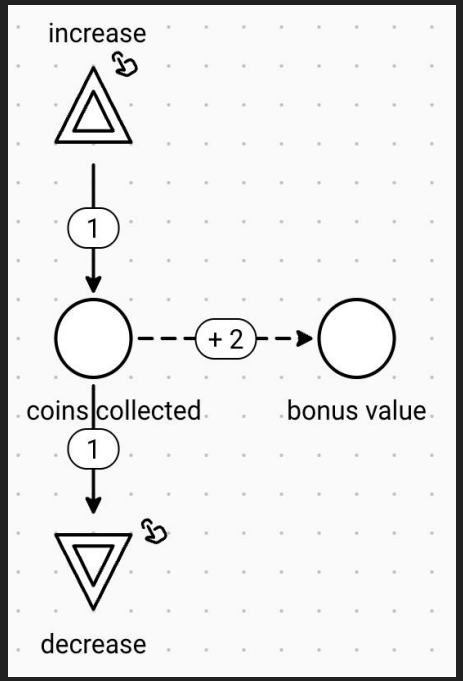
For example, from an analysis of countless idle games and clickers, a 'magic number' has emerged for the cost increase of upgrades. This is the ideal ratio to keep the game challenging yet achievable.

The Sweet Spot: A ratio between 1.05 and 1.20 is most effective. This number creates a noticeable cost increase without making the player feel like the game becomes impossible.

Example: If an upgrade costs 100 with a ratio of 1.15, the next one will cost 115, the one after that 132, and so on.



Example: State connection



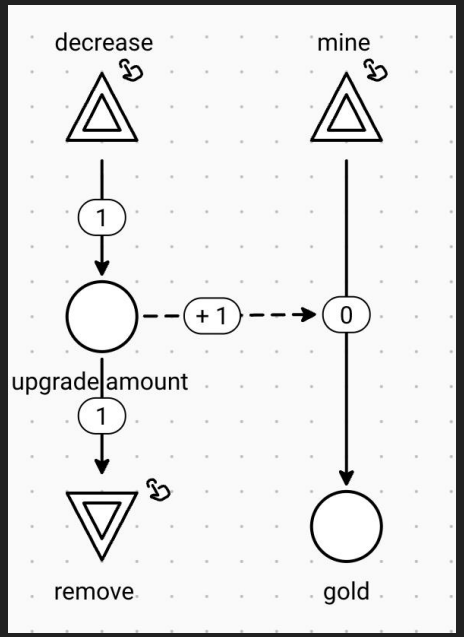
This diagram shows how collected coins can automatically determine a bonus value.

The left pool represents ‘Coins Collected,’ which you can increase or decrease. The right pool, ‘Bonus Value,’ is controlled by a state connection: it always receives twice the amount of the left pool.

This models a game mechanic where rewards scale with what the player collects.

The diagram demonstrates how you can use state connections to modify a resource Pool.

Example: Modify connection value



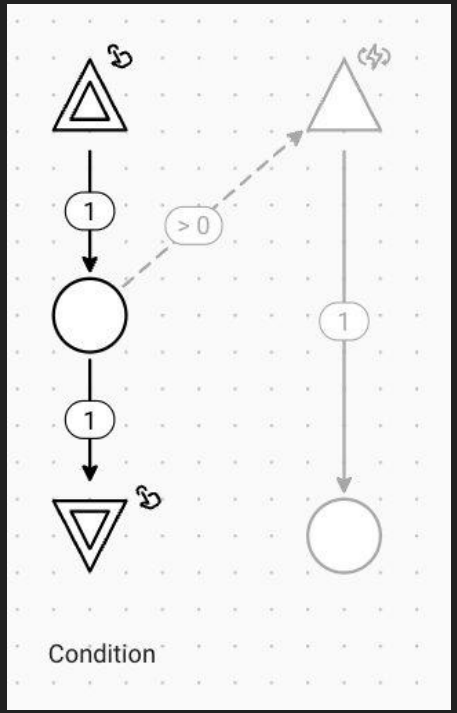
This diagram shows a simple setup with a left Pool and a Source/Drain that the player can interact with to increase or decrease the Pool.

Unlike the previous example, the right side is a connection rather than a Pool. The weight of this connection is determined by the amount in the left Pool.

In a game context, this models a tool upgrade mechanic in a simple mining game: each +1 in the left Pool increases the connection's weight by +1, effectively adding +1 gold per mining action.

The diagram demonstrates how a state connection can dynamically adjust its effect based on a resource Pool, allowing upgrades or modifiers to scale with player progress.

Example: Condition

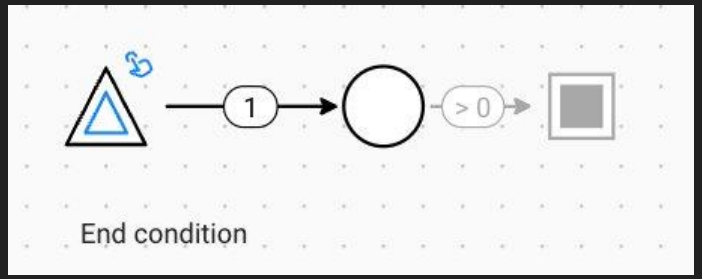


This diagram demonstrates a powerful concept in game design: the condition. A condition is a rule that must be met for an action to occur.

In this example, the player can use the Condition action on the left to activate an output. This output, however, is not immediately available. The dotted line with the > 0 icon shows that the mine action on the right can only be performed if the number in the Condition pool is greater than zero.

This simple rule is a powerful tool. It allows you to model abilities that can only be used a limited number of times, or an action that is only unlocked once the player has reached a certain milestone. This diagram shows how you can use conditions to control and direct player choices.

Example: End Condition

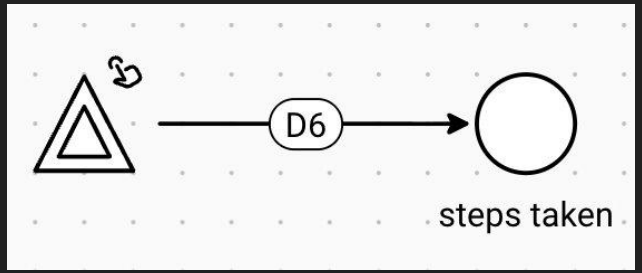


This diagram demonstrates how you can visualize the end of a game. In a progression game, the game ends when a player reaches a specific milestone. This milestone is called the end condition.

In this example, the End Condition node on the left represents an action that, when performed, adds a resource to a pool. The game is over once the amount in that pool becomes greater than zero.

This simple rule is a powerful tool. It allows you to model game-ending goals like gathering a certain number of points, unlocking a final upgrade, or reaching a specific score.

Example: The Dice Roll

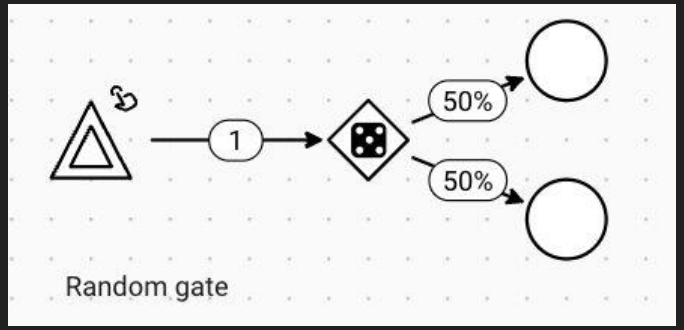


This diagram demonstrates how to visualize a key element of tabletop games: randomness. This is one of the ways to add variety and unpredictability to your game.

In this example, the Dice Roll node represents a six-sided die (D6) that produces a random number from 1 to 6. The result of this roll determines the number of steps taken by the player's pawn.

This diagram is a simple model for many random mechanics, such as a gambling system, an unpredictable event, or the movement of a pawn. It shows you how to integrate an element of chance into your game's design.

Example: The Random Gate

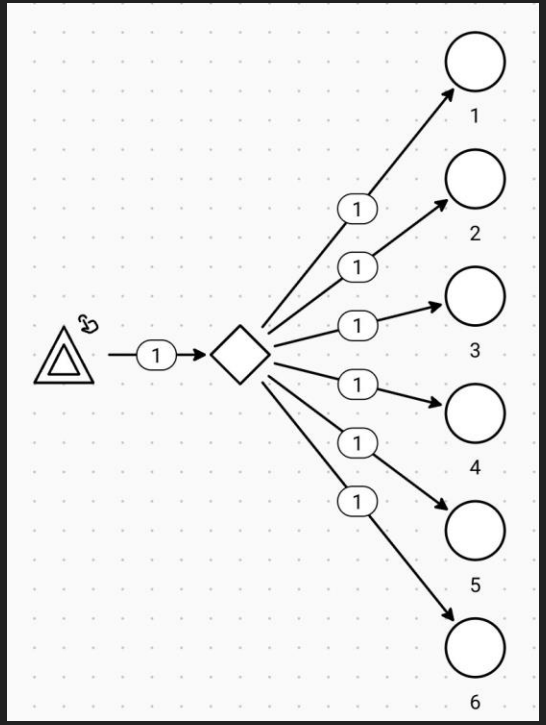


This diagram shows you how to model a 50/50 chance, a common mechanic in games of luck and gambling.

The central Random Gate node acts like a digital coin flip. A resource that enters this gate has an equal chance (50%) of going down one of two paths. In a game context, this can represent a simple gamble, like opening a box that might contain a reward.

The diagram demonstrates how to create a simple, unpredictable outcome. This is a crucial tool for adding randomness and risk to your game, which can make a strategy-focused game feel more exciting and replayable.

Example: The D6 Roll

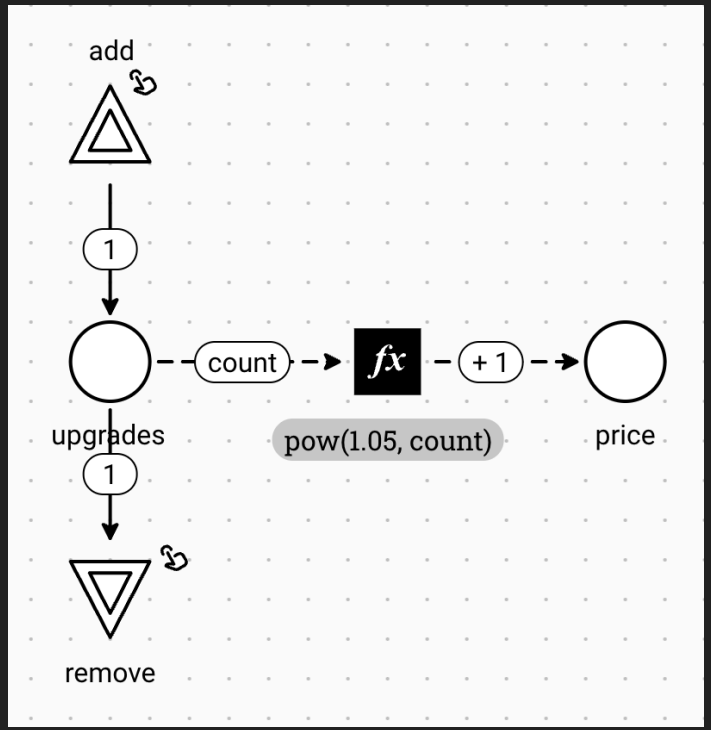


This diagram shows you how to model the full range of outcomes of a standard six-sided die (D6).

The central Random Gate node acts as a six-sided die. A resource entering this gate will randomly exit through one of the six paths, each representing a number from 1 to 6. This is different from the previous examples, as it visualizes every possible outcome and its corresponding path.

This is a key tool for creating mechanics where the result of a random event has a direct and variable impact on the game, such as determining movement, a variable resource gain, or a specific event trigger. It shows you how to integrate a direct element of chance into your game's design.

Example: Exponential Growth

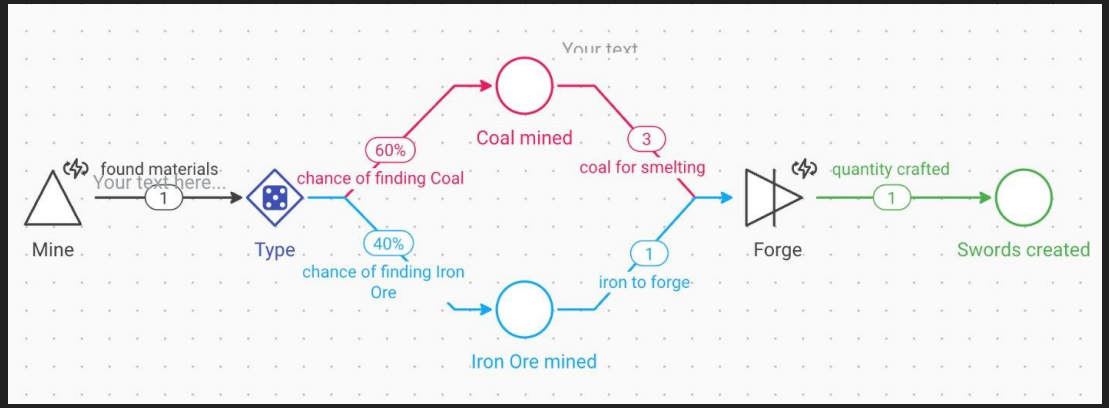


This diagram shows you how to model exponential growth, a core mechanic in many progression games.

The upgrades pool keeps a running count of how many times a player has upgraded. This count is then used in the formula pow(1.05, count) to calculate the price of the next upgrade. This means that each new upgrade will cost 1.05 times the previous one, leading to an exponentially increasing price.

This is a key tool for creating mechanics where the price or power of an item scales dramatically as the game progresses. It is the engine that drives the feeling of progress in a game and encourages players to make strategic choices.

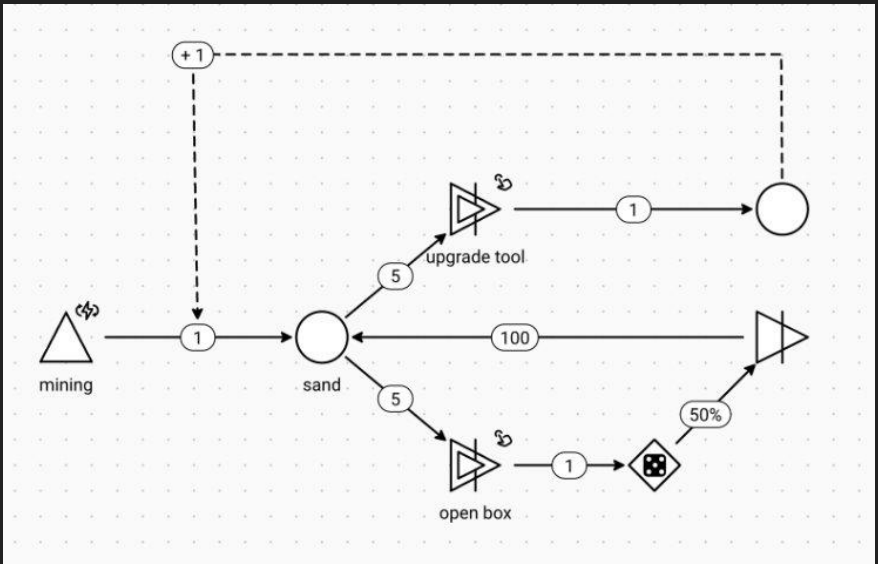
Example: Crafting



This diagram shows a layered production pipeline, where you gather raw materials to then craft a final product. The player Mines, and a random gate determines what material they find: Coal (60% chance) or Iron Ore (40% chance). Both are needed for the next action.

This diagram demonstrates the essence of a crafting system in a game. You gather a variety of materials and then process them into a finished product. This can be used to model a player's progression, or to create a complex economy.

Machinations Diagram: A Core Loop



This diagram demonstrates the core economic loop of a simple mining game. Sand is the primary resource in the game. The player starts with a mining action, which generates sand.

Currently, the sand can be spent in two ways:

- Upgrade Tool: The player can spend 5 units of sand to buy an upgrade. This upgrade permanently increases the amount of sand they earn with each mining action. This is the progression loop.
- Open Box: The player can spend 5 units of sand to open a box. There is a 50% chance to gain a bonus of 100, and a 50% chance to gain nothing. This is the gambling loop.

This diagram serves as a foundational example for your assignment. It
shows you how to model the core loop of a progression game and how
to model options for the player.

Acknowledgement

These slides were based on the excellent lecture of Yvens Rebouças Serpa, author of the book "The cores of game design"
If you want to read more, here is a link to the original slides and his site.

